9TH YEAR

#NATIONAL TECHNOLOGY INITIATIVE

INTERNATIONAL UNMANNED AERIAL VEHICLES COMPETITION

FREE MISSION CATEGORY

PROJECT PRESENTATION REPORT

ASTREUS TEAM



INTERNATIONAL
UNMANNED AERIAL VEHICLES COMPETITION
FREE MISSION CATEGORY
PROJECT PRESENTATION REPORT

TEAM NAME: ASTREUS TEAM
TEAM TYPE: MIXED TEAM
VEHICLE TYPE: VTOL
VEHICLE DEVELOPMENT TYPE: NEW VEHICLE
UNIVERSITY NAME: Aristotle University of Thessaloniki / University of Western Macedonia / University of Thessaly
COUNTRY: Greece
TEAM RESPONSIBLE: Vasileios Theofylaktos

1. SUMMARY

1.1. Team Organization

Project Sponsorship

Our team operates as an independent, mixed-university student initiative. The development, and procurement of our VTOL platform are sponsored by the Municipality of Voio, Greece.

Role Distribution & Responsibilities

- **Team Responsible / Member 1 / Mechanical Lead**

Rensposible for the Project timeline, assignment of tasks within the team and communication with our sponsor, Municipality of Voio. Also Leads Mechanical engineering, wing and tail design and manufacture, aerodynamic and cg analysis.

- **Member 2 – Avionics Lead / Software Lead**

Responsible for selection of avionics components (Power System, Propulsion System, Power Regulation and Distribution, Control Surfaces, Sensors), electronics simulations on eCalc and overall electronic circuitry. Also responsible for the Drone Software.

- **Member 3 – Mechanical Engineer**

Responsible for the Ailerons (for uav's turns) design and manufacture. Also responsible for the overall manufacturing and material selection for the uav's wings and fuselage.

- **Member 4 – Telemetry Lead**

Responsible for the communication with the Drone. Telemetry design, simulations and manufacturing.

- **Member 5 – Ground Station Engineer**

Responsible for hardware design and manufacturing of the antenna tracking platform at the ground station.

- **Member 6 - Ground Station Engineer**

Responsible for Software development of the antenna tracking platform at the ground station.

1.2. Mission Title

Mission Name: Autonomous VTOL Medical Transportation: Proof of Concept.

General Framework & Objective: The mountainous region of Voio features isolated villages without direct access to medical supplies. The main objective of this mission is to demonstrate a Proof of Concept (PoC) for our 17.5 kg MTOW hybrid VTOL QuadPlane. This flight will strictly validate the core autonomous transitions, the aerodynamic stability during a 2 KG payload precision airdrop, and our antenna tracking platform designed to maintain optimal alignment between the ground antenna and the UAV. Also, by utilizing a custom-built ground base platform to catch the UAV mid-air, we eliminate the need for traditional landing legs, while optimizing the aerodynamic flow of the vtol fixed wing uav.

Sequential Demonstration Flow: During the field demonstration, the UAV will perform the following actions in a strictly sequential flow:

1. Pre-flight & AAT Link: Final avionics initialization and verification of the telemetry link via our custom Automated Antenna Tracker (AAT) on the launch platmform.
2. Vertical Ascent: Autonomous multirotor takeoff to a predetermined safe transition altitude.
3. Forward Transition: Autonomous engagement of the pusher motor and sequential shutdown of the VTOL lifting motors to enter aerodynamic flight.
4. Cruise Navigation: Fixed-wing autonomous circuit flight, navigating through pre-programmed MAVLink waypoints.
5. Mission Execution (PoC Airdrop): The UAV transitions back to hover mode over the designated drop zone, descending to a safe altitude to execute the release of the 2 KG payload.
6. Return & Reverse Transition: The UAV climbs, transitions back to fixed-wing flight, and returns to the home coordinate.
7. "Zero-Touch" Aerial Landing: The UAV executes a precision multirotor descent directly into our custom-built ground recovery base, where mechanical arms physically secure the reinforced composite wings mid-air, concluding the mission without ground contact.

2. MISSION INFORMATION

2.1. Flight Mission Description

Purpose, Scope, and Objectives

The primary purpose of this mission is to execute a Proof of Concept (PoC) for the Astreus 17.5 kg MTOW hybrid VTOL QuadPlane. The scope encompasses validating the aerodynamic stability of the heavy-lift transitions, executing a precision 2 KG payload airdrop, and demonstrating our DIY ground station platform for the landing and our DIY antenna tracking

platform. The objective is to simulate a remote medical delivery cycle that's completely automated, reliable and fast in the mountainous region of Voio.

Flight Scenario (Initiation, Execution, and Completion)

The mission unfolds as a strict, sequential scenario:

- **Initiation**: The Ground Control Station (GCS) establishes a solid MAVLink connection via our custom Automated Antenna Tracker (AAT) and the onboard RFD 868x-EU telemetry module. The Cube Orange is armed on the launch pad.

- **Execution**: The UAV performs a vertical takeoff to a safe altitude for the mission. It executes the forward transition: the pusher motor engages, the airspeed sensor confirms stall-safe velocity, and the VTOL motors shut down. The UAV enters fixed-wing cruise, navigating through pre-programmed MAVLink waypoints.

- **Target Phase**: Upon reaching the drop coordinates, the UAV executes a reverse transition into a stable hover at a 5-meter safety altitude. The automated bay opens, releasing the 2 KG payload.

- **Completion ("Zero-Touch" Landing):** The UAV climbs, transitions back to fixed-wing flight, and returns to the home coordinate. Instead of a traditional touchdown, the UAV executes a precision multirotor descent directly into our custom-built ground recovery base, where mechanical arms physically secure the reinforced composite wings mid-air.

Difficulty Level, Success Criteria, and Performance Metrics

- **Difficulty Level**: Hard. Managing the transition of a 17.5 kg airframe is complex, also the manufacturing of a reliable tracking system for our DIY antenna to communicate with the uav is a crucial challenge.

- **Success Criteria**: (1) Successful cruise transition without exceeding a 3-meter altitude drop. (2) Payload delivery within a 5-meter radius. (3) Flawless telemetry (without data loss) and mid-air capture by the ground base mechanism.

- **Performance Metrics:** AAT tracking accuracy (Signal-to-Noise Ratio retention), CG (Center of Gravity) compensation time after the 2 KG drop, and precision landing deviation (in centimeters) from the ground recovery platform.

Autonomy and Manual Intervention Limits

The mission is 100% autonomous. The precision required for the mid-air catch vastly exceeds human piloting capabilities; thus, manual landing is not an option. Decision-making is governed entirely by the ArduPilot firmware. Manual intervention is strictly restricted to emergency abort procedures.

System Response to Changing Conditions & Fail-Safes

Mission continuity is ensured through multi-layered EKF3 automated responses:

- **Signal Loss:** If the custom AAT mechanism fails to track the UAV and the MAVLink is lost for more than 5 seconds, the UAV enters an automatic Return-to-Launch (RTL) protocol to regain signal line-of-sight.
- **Wind and Obstacles:** During cruise, the DroneCAN DLVR airspeed sensor continuously monitors True Airspeed. Against headwinds, the autopilot automatically increases pusher throttle to prevent stalling.

- **GPS Errors:** In the event of GNSS denial or severe satellite degradation, executing the precision aerial catch becomes impossible. The system will fall back to the Cube Orange's triple-redundant IMUs, abort the precision landing, and trigger a fail-safe Q-Land sequence to descend immediately and safely away from the mechanical base.

2.2. Flight Mission Originality

Operational Originality: Overcoming Topographical Bottlenecks The mission's originality lies in addressing the severe topographical limitations of the Voio Municipality. Traditional multicopters lack the endurance for remote mountain deliveries, and fixed-wing UAVs cannot operate without runways. Our approach creates a Rapid-Response long range Medical Delivery network using a 17.5 kg MTOW hybrid VTOL QuadPlane, requiring absolutely zero runway infrastructure while maintaining high-speed cruise efficiency.

Innovations

Antenna tracking platform

It is crucial for our team to design and manufacture a reliable tracking system so we can maintain a stable communication with our uav. Since the system is fully autonomous and we plan a Beyond Visual Line of Sight, it is essential that we know the live location, battery percentage and the crucial temperatures in the ground station.

The "Zero-Touch" Aerial Recovery System

A major challenge in pusher-VTOL platforms is protecting the rear 16-inch propeller during vertical landings without using heavy, drag-inducing extended landing gears while also increasing the medical payload capacity.

3. UAV INFORMATION

3.1. Design

Overall Design Approach & Configuration

The Astreus UAV is designed specifically to address the demanding mission requirements of long-range, high-payload medical delivery in mountainous terrain. The selected configuration is a Hybrid VTOL QuadPlane. This design natively combines the runway-independent vertical

deployment capabilities of a multirotor with the aerodynamic efficiency and high-speed cruise stability of a fixed-wing aircraft.

Dimensions, Weight, and Propulsion System

- **Dimensions:** The airframe has a wingspan of 3000 mm and a fuselage length of 1200 mm.
- **Take-off Weight:** The Maximum Take-Off Weight (MTOW) is strictly limited to 17.5 kg , which includes the 2 kg medical payload.
- **Propulsion:** A dual-system architecture is implemented. The VTOL phase is powered by four high-torque vertical lifting motors (T-Motor V605). The forward cruise phase is powered by a single rear pusher motor (T-Motor AT5230A).

Take-off and Landing Architecture (Zero-Touch Base)

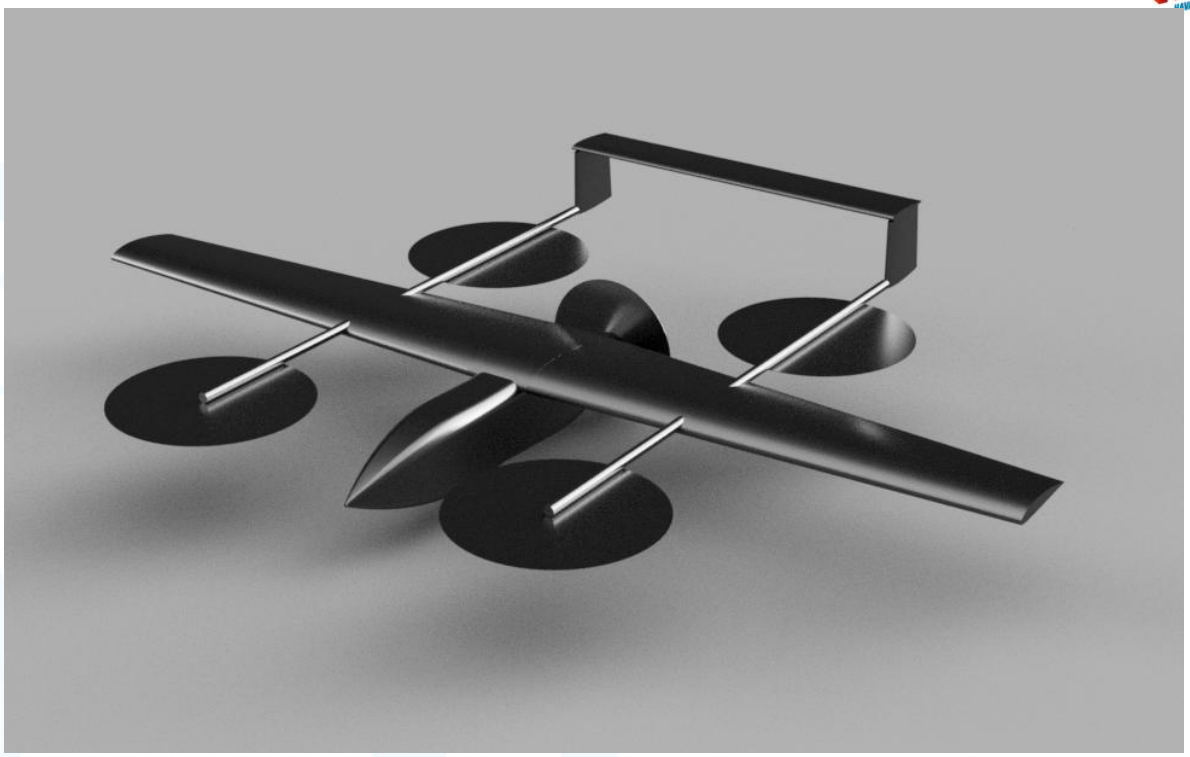
In a radical departure from conventional designs, the UAV features no traditional landing gear (such as wheels, skids, or parachutes).

- **Take-off:** Executed vertically from a standard flat ground pad.
- **Landing:** Executed via the same ground platform as the take-off. Also includes a crucial challenge: we have to secure the automatic landing to be precise in the platform to avoid damaging our uav.

Flight Stability Analysis

- **Hover Stability:** During take-off and landing, stability is actively maintained by the Cube Orange flight controller utilizing dynamic thrust vectoring across the four V605 motors, compensating for crosswinds and ground-effect turbulence.
- **Cruise Stability:** The aerodynamic profile is optimized for a stable cruise at 25m/s. The Center of Gravity (CG) is meticulously calculated to remain neutral even after the 2 kg payload is dropped.
- **Transition Stability:** The critical transition from hover to cruise is heavily stabilized by real-time True Airspeed monitoring (via a DroneCAN pitot tube), ensuring the VTOL motors only disengage when the fixed-wing surfaces generate sufficient lift, preventing stalls.
- **Ailerons:** To ensure level-flight stability and mission precision, the composite ailerons utilize rigid, zero-backlash linkages. This durable configuration prevents aeroelastic flutter, maximizing control authority during the critical airdrop phase.

Drone CAD



3.2. Hardware

Hardware Architecture and System Description

The entire system architecture of the UAV is centered around a robust power distribution and avionics infrastructure designed to support both VTOL and fixed-wing flight modes while ensuring safe and reliable operation.

Power System

The power source of the UAV consists of two GENS ACE Tattu Semisolid State 350Wh/kg 6S 33000mAh 10C XT90S LiPo batteries connected in series. Each battery is rated at 6S, 33000 mAh and 10C, resulting in a combined 12S system voltage, which is required for the efficient operation of the motors.Due to the high current requirements of the motors, power Distribution board (PDB) is not a good selection for our system. Instead, the system uses high-current wiring (AWG7–AWG8) routed along the side sections of the drone structure. These cables distribute the battery voltage directly to all components that require high-power input.

Propulsion System

The UAV features a hybrid propulsion system that supports both VTOL operation and forward flight. For vertical takeoff and landing, the aircraft is equipped with four VTOL motors:V605 KV210 12S VTOL Brushless Drone Motor Each motor is controlled by a electronic speed controller: FLAME 100A 14S MultiRotor UAV ESC. To reduce electrical losses and improve efficiency, the ESCs are positioned close to the battery. The power cables connecting the ESCs to the battery are approximately 100 mm long, while the cables connecting the ESCs to the motors extend up to 1 meter, allowing proper routing along the UAV arms.

For forward thrust during fixed-wing flight, the UAV includes a propulsion

motor: AT5230A 30CC VTOL Fixed Wing Aircraft Long Shaft Motor KV200 This motor is controlled by the corresponding ESC: AT 115A 614S VTOL Fixed Wing ESC. This propulsion configuration allows the UAV to perform vertical takeoff and landing while transitioning to efficient forward flight, significantly extending mission endurance.

Power Regulation and Distribution

Power lines from the battery continue inside the UAV fuselage where they supply several DC-DC converters responsible for generating the required voltage levels for avionics and payload systems.

The system includes:

- Five DC-DC converters reducing voltage to 7.5 V
- Two DC-DC converters reducing voltage to 5 V
- One voltage regulator generating 3.3 V from the 5 V rail

The 7.5 V converters supply the high-current servo bus. Their outputs are combined through diode-based current sharing, forming a single power node that directly feeds the servos. This architecture provides redundancy and prevents reverse current flow between converters.
The 5 V converters serve two separate purposes:

- One dedicated converter powers the flight controller independently.
- The second supplies all other 5 V avionics components.

The 3.3 V regulator is used exclusively for the Bluetooth communication module. This multi stage regulation architecture ensures stable power delivery to all subsystems while isolating critical avionics from potential voltage fluctuations.

Control Surfaces and Actuation

The aircraft uses a total of eight servo motors:

- Six high-torque servos (35 kg·cm) controlling the aircraft control surfaces (ailerons and other aerodynamic control mechanisms)
- Two servos (11 kg·cm) responsible for the payload release mechanism

This configuration ensures precise control authority during both VTOL and fixed-wing flight phases.

Avionics and Sensors

The UAV avionics are centered around the Cube Orange+ flight controller, which performs the main processing of flight data, navigation, and control commands.

Several sensors and communication modules are directly connected to the flight controller:

- HM10 Bluetooth Module for Arduino

Used to transmit mission data (such as destination coordinates) from a mobile application directly to the flight controller.

- Gravity URM09 Ultrasonic Distance Sensor (2–500 cm, I2C)

Used to assist with landing by measuring altitude relative to the ground during the final phase of descent.

- CubePilot Here+ RTK GPS

Provides high-precision positioning data required for navigation and mission execution.

All sensors communicate directly with the flight controller, ensuring centralized processing and improved system reliability.

System Integration and Safety Considerations

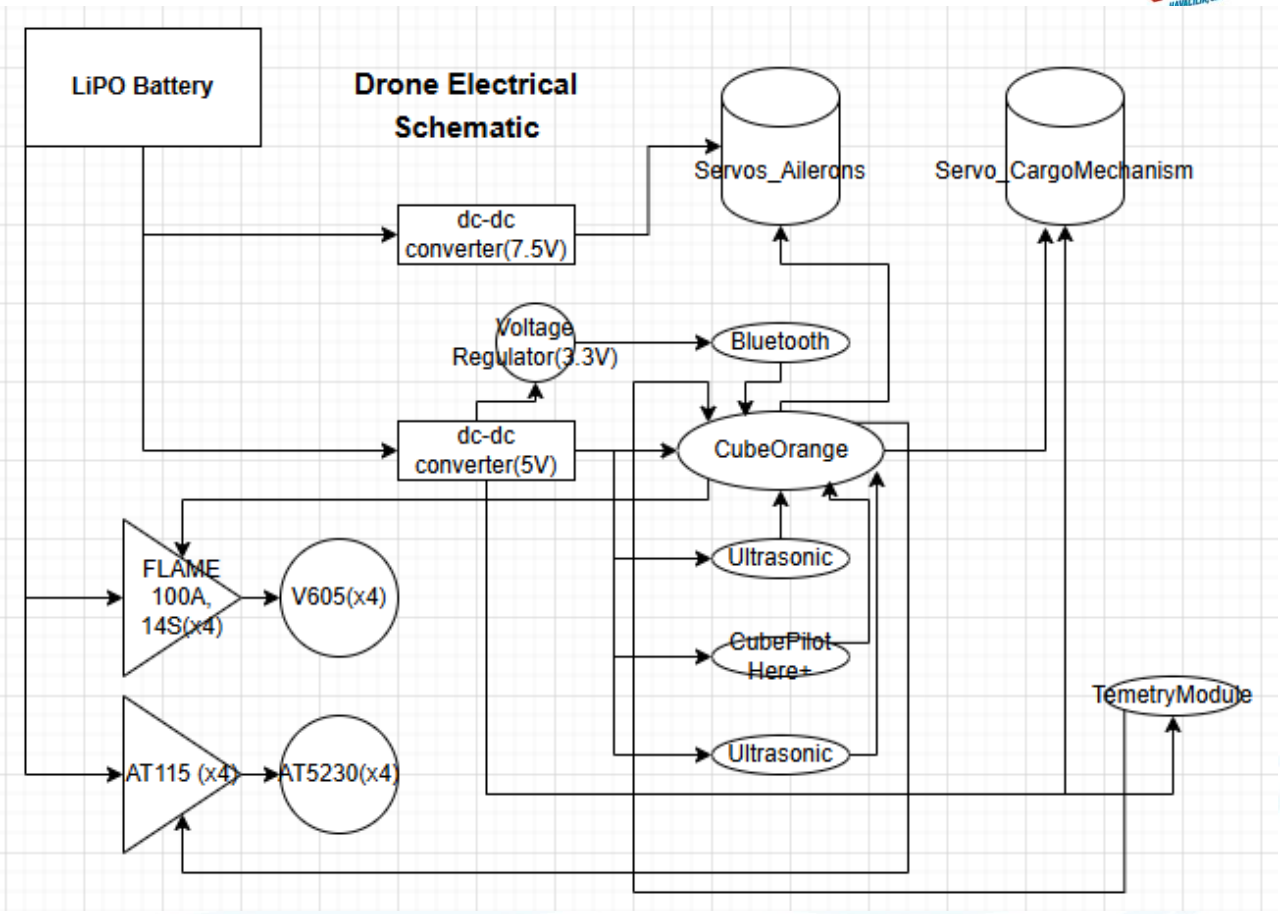
All the components of the system are connected through properly planned wiring and electrical connections to make sure they communicate reliably with the flight controller. The Cube Orange+ works as the main control unit of the UAV. It receives data from the sensors, processes the flight algorithms, and sends commands to the propulsion system and the control surfaces.

The hardware design avoids unnecessary or extra components and focuses mainly on reliability and efficient power use. High-current cables are kept as short as possible, and different voltage levels are provided separately for different subsystems. Important avionics components are also powered independently. This helps improve flight safety and reduces the chance of a failure affecting the entire system.

Telemetry

For the telemetry section, except of using the Hex Here3+ GPS module, we decided to use the RFD 868x-EU Telemetry module, it's set up at 868MHz, one of the most common frequencies in the drone world. The module itself has a power of about 0.5watt and with the use of two ProFlex_R Dipole 868MHz antennas we aim to accomplish the distance of 20km. As for the ground station, we will also use the same RFD868x-EU module, but with the use of a Yagi 868Mhz antenna made by our team. For the construction of the antenna a pvc tube, as the main body (Boom), will be used. The Grid Reflector will be constructed by a metal mesh, with dimensions 20x20cm and mesh opening under 3.5cm. The Elements(Driven Element and Directors) will be made from a solid copper, aluminum, or brass wire, with dimensions 162mm, the Driven Element, 152mm, 149mm, 146mm, 143mm, the 4 Directors, and with diameter around 2-3mm. Finally, the antenna will connect with the module by a 50Ohm Coaxial Cable (LMR-400) and a RM-SMA(Reverse Polarity SMA) male connector.

Additionally, at the ground station we plan to integrate a mechanical antenna tracking platform designed to maintain optimal alignment between the ground antenna and the UAV during flight. The system is inspired by the positioning mechanisms used in astronomical telescopes. The Yagi antenna will be mounted on a two-axis rotating structure capable of controlling both azimuth and elevation. The movement of the platform will be achieved using two high-torque motors, each responsible for one rotational axis, allowing precise orientation adjustments of the antenna. The mechanical structure will be designed to provide sufficient rigidity and stability so that the antenna maintains accurate pointing during operation, even under environmental disturbances such as wind. This hardware configuration enables the ground station antenna to dynamically adjust its direction toward the UAV, improving telemetry link quality and communication reliability over long distances.



3.3. Software

Software Architecture and Mission Execution

The software architecture of the UAV system has three main components: the onboard flight control software, the ground station control system, and the mobile application interface used to initialize and monitor the mission.

The UAV flight controller is a Cube Orange+, which runs the ArduPilot autopilot firmware. The firmware is uploaded to the flight controller using the Mission Planner ground control software, which also allows configuration and system monitoring before the mission begins.

Mission initialization begins through a mobile application, from which the operator defines the destination coordinates of the UAV. These coordinates are transmitted wirelessly to the UAV using the onboard HM10 Bluetooth Module for Arduino. The coordinates are then stored as mission parameters that define the navigation target of the aircraft.

Alongside the main autopilot firmware, the Cube Orange+ executes a Lua script through the ArduPilot scripting engine. This script defines the high-level mission logic and controls the sequence of operations that the UAV performs during the flight.

The mission is organized into several operational phases.

Takeoff Phase

At the beginning of the mission, the UAV performs a vertical takeoff using its four VTOL propulsion motors. During this phase, the flight controller continuously monitors altitude, velocity, and position using onboard sensor data. The aircraft climbs until it reaches a predefined safe altitude. Throughout this phase, telemetry data such as position, speed, and system status are transmitted to the ground station.

Transition and Acceleration Phase

Once the target altitude is reached, the UAV transitions from vertical flight to forward flight. The fifth propulsion motor responsible for forward thrust is activated and the aircraft begins accelerating.The system attempts to reach a cruise speed of approximately 90 km/h, a value determined through propulsion and endurance simulations performed using the eCalc analysis tool. This cruise speed ensures that the mission can be completed within the time constraints while maintaining efficient energy consumption.

Navigation Phase

During the cruise phase, the UAV navigates toward the target location using positioning data obtained from the CubePilot Here+ RTK GPS module. The flight controller continuously compares the current position of the aircraft with the target coordinates and performs trajectory corrections by adjusting the control surfaces of the aircraft, primarily the ailerons. These corrections allow the UAV to maintain the optimal trajectory toward the destination.

Delivery Phase

When the UAV enters a predefined arrival radius around the target waypoint, the landing sequence begins. During descent, the aircraft uses the Gravity URM09 Ultrasonic Distance Sensor to measure the distance from the ground and ensure a controlled landing. Once the measured altitude falls below approximately 0.5 meters, the UAV stabilizes its position and activates the payload release mechanism, delivering the medical supplies to the designated location.

Return Phase

After the payload has been released, the UAV automatically initiates the return-to-base procedure. The same navigation algorithm is executed again, but this time the target waypoint corresponds to the original takeoff location.

Final Landing

Upon arrival at the launch location, the UAV performs a controlled landing procedure similar to the delivery landing phase, but without activating the payload release system.

Ground Station Software

The ground station software is responsible for telemetry reception, antenna tracking control, data logging, and real-time mission monitoring.

At the beginning of the mission, the ground station knows the initial coordinates of the UAV launch location. Using this information, the antenna tracking system is initially oriented toward the expected position of the aircraft in order to receive the first telemetry signals after takeoff.

Telemetry data transmitted by the UAV are received through the RFD 868x-EU Telemetry Module operating at 868 MHz. The received data include flight parameters such as position, speed, altitude, system status, and mission state.

The ground station software continuously processes the incoming telemetry stream and calculates the relative position of the UAV with respect to the ground station. These calculations are used to control a mechanical antenna tracking system that maintains the directional antenna aligned with the aircraft.

The antenna is mounted on a two-axis rotating platform, which controls both azimuth and elevation. Two motors adjust the orientation of the antenna so that it continuously follows the UAV during the entire mission, including the return phase.

At the same time, a parallel software process continuously logs all incoming telemetry data on the ground station computer for monitoring.

The telemetry stream is then forwarded to the operator's mobile application using a MAVLink telemetry relay over a local Wi-Fi network. In this architecture, the ground station computer acts as a telemetry server, allowing the mobile application that initiated the mission to display real-time flight data, UAV position, and mission progress.

3.4. Originality-Team Production and Domestic Product Use

To maximize our engineering capabilities and minimize reliance on expensive Commercial Off-The-Shelf (COTS) products, the Astreus team has heavily invested in in-house manufacturing, spanning advanced composites, robotics, and RF engineering.

Mechanical Production: Custom Composite Airframe & Zero-Touch Recovery Base We entirely avoided pre-manufactured foam airframes. The complete VTOL fuselage and lifting surfaces are natively designed and produced by our team. The wings are constructed utilizing custom aerodynamic profiles optimized for the 17.5 kg MTOW, manufactured through in-house composite layup techniques (carbon fiber spars integrated with a fiberglass/epoxy skin). This ensures maximum structural integrity, while the XPS foam helps minimize the drone's weight in the wings and the fusage. Furthermore, the Zero-Touch aerial Recovery platform is also team-produced.

Mechatronics & RF Production: High-Gain Yagi & Automated Antenna Tracker (AAT) In the communications sector, we designed a fully custom BVLOS telemetry architecture. Our RF engineers are going to calculated, cut, and assembled a highly directional Yagi antenna strictly tuned for the 868MHz band to maximize the Signal-to-Noise Ratio (SNR). Additionally, our team will built an Automated Antenna Tracker (AAT) system. Because the Yagi antenna has a narrow beamwidth, we'll avoid manual aiming. To solve this, our mechatronics team is developing a custom Pan-Tilt targeting mechanism governed by a dedicated microcontroller. By actively parsing real-time MAVLink GPS telemetry from the UAV, the AAT dynamically rotates and elevates the heavy-duty Yagi antenna to maintain a perfect lock on the moving aircraft.

4. COST ANALYSIS AND BUDGET TABLE

No	Materials / Services Planned to be Purchased with Project Support (Brand and Model information must be written)	Piece	Unit Price (in TRY)	Total Amount (in TRY)	Reason for Purchasing Material / Service
1	Traveling and accommodation costs	-	-	120000	For the transportation (from Greece) and accommodation for our 6 member Team where the competition will be held.
TOTAL PROJECT SUPPORT REQUESTED FROM TÜBİTAK*				**120000 TRY**	

* Project Support upper limit is 120,000 TRY.

** Rows can be added to the table if necessary. The format of the table should not be changed.

5. REPORT LAYOUT

In this section, the format requirements of the Project Presentation Report will be evaluated. In order to create a fair evaluation environment for the teams, some limitations have been imposed on the number of pages and spelling rules.

- The report should be written in accordance with the explanations provided in the Guide for Project Presentation Report prepared for the Free Mission category. The format of the report template must not be changed.
- Font should be set to **Arial**, Point to **11**, Line Spacing to **1.15**, including tables and formulas.
- Page layout should be **A4 type** and margins should be **Top 3 cm; Right, Left and Bottom 2.5 cm**. The **page** should be used **vertically** (Workflow chart and circuit diagram pages can be used horizontally).
- Project Presentation Report should not exceed 17 pages, including the TEKNOFEST covers at the beginning and end of the template and the appendices.

- The TEKNOFEST covers at the beginning and end of the template must not be deleted. It should be noted that if teams create their own cover pages, the relevant cover is counted as 1 page.
- Figure sizes, information, and details should be readable without the need for any magnification.
- All text must be aligned on both sides (justified).
- Report pages should be numbered consecutively in accordance with the format.
- The team name and page number should be written in the footer of the pages.
- All sources used in the report should be stated.
- It is expected that the report includes the innovations carried out and planned by the teams in their own words, rather than general information.

  Example 1: *If a UAV is designed for firefighting, the unique aspects of the team's work should be emphasized rather than general information such as fire statistics, the importance of firefighting, or loss of life and property.*

  Example 2: *In the case of dropping first aid supplies by UAV, it is more important to explain how this is performed.*
- The report should be uploaded to KYS **in PDF format**.
- In case of detection of ethical violations such as plagiarism or copying, including any elements not specific to the team (such as other teams' mission information or design visuals), the report will not be evaluated and the team will be eliminated from the competition. The team cannot object to this situation.

- **FREE MISSION CATEGORY PROJECT PRESENTATION REPORT EVALUATION SCORES**
- Project Presentation Reports are scored according to the table below.

No	Section	Point
1	Team Organization	**4**
2	Mission Title	**4**
3	Flight Mission Description	**18**
4	Flight Mission Originality	**10**
5	Design	**10**
6	Hardware	**10**
7	Software	**10**
8	Originality-Team Production and Domestic Product Use	**24**
9	Cost Analysis and Budget Table	**5**
10	Report Layout	**5**

9TH YEAR

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